

4.1 VECTOR SPACES AND SUBSPACES

Def: A vector space is a non-empty set V of objects (vectors) on which are defined 2 operations:
 $+$ addition
 $\cdot$ multiplication.

with 10 axioms.

$$1^{\circ} \text{ if } \bar{v}, \bar{u} \in V \text{ then } \bar{u} + \bar{v} \in V$$

$$2^{\circ} \bar{u} + \bar{v} = \bar{v} + \bar{u} \quad \forall \bar{u}, \bar{v} \in V$$

$$3^{\circ} (\bar{u} + \bar{v}) + \bar{w} = \bar{u} + (\bar{v} + \bar{w}) \quad \forall \bar{u}, \bar{v}, \bar{w} \in V$$

$$4^{\circ} \exists \bar{0} \in V: \bar{u} + \bar{0} = \bar{0} + \bar{u} = \bar{u} \quad \forall \bar{u} \in V$$

$$5^{\circ} \forall \bar{u} \in V \exists -\bar{u} \in V: \bar{u} + (-\bar{u}) = \bar{0}$$

$$6^{\circ} \forall \bar{u} \in V, \forall k \in \mathbb{R} \quad k \cdot \bar{u} \in V$$

$$7^{\circ} k(\bar{u} + \bar{v}) = k\bar{u} + k\bar{v} \quad \forall k \in \mathbb{R}, \forall \bar{u}, \bar{v} \in V$$

$$8^{\circ} (k+l) \cdot \bar{u} = k \cdot \bar{u} + l \cdot \bar{u} \quad \forall k, l \in \mathbb{R}, \forall \bar{u} \in V$$

$$9^{\circ} k \cdot (l \cdot \bar{u}) = (k \cdot l) \cdot \bar{u} \quad \forall k, l \in \mathbb{R}, \forall \bar{u} \in V$$

$$10^{\circ} 1 \cdot \bar{u} = \bar{u} \cdot 1 = \bar{u} \quad \forall \bar{u} \in V$$

Ex 1 $\mathbb{R}^n$ - is a premier example of a VS

Ex 2 Set of all infinite sequences (S_n) - is a VS.

$$\{y_k\} = \{\dots, y_{-2}, y_{-1}, y_0, y_1, y_2, \dots\}$$

$$(\oplus): \{y_k\} + \{z_k\} = \{\dots, y_{-2} + z_{-2}, y_{-1} + z_{-1}, y_0 + z_0, y_1 + z_1, y_2 + z_2, \dots\}$$

$$(\otimes): k \cdot \{y_k\} = \{\dots, k \cdot y_{-2}, k \cdot y_{-1}, k \cdot y_0, k \cdot y_1, \dots\}$$

Ex 3 Set of all polynomials of degree "n" (P_n) - is a VS.

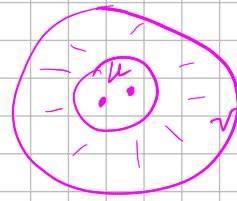
$$p(t) = p_0 + p_1 t + p_2 t^2 + \dots + p_n t^n$$

$$(\oplus) (p+q)(t) = p_0 + q_0 + (p_1 + q_1)t + (p_2 + q_2)t^2 + \dots + (p_n + q_n)t^n$$

$$(\otimes) k \cdot p(t) = k p_0 + k p_1 t + k p_2 t^2 + \dots + k p_n t^n$$

Subspaces

$$U \subseteq V$$



$$1^\circ \quad \vec{0} \in V \Rightarrow \vec{0} \in U \quad \checkmark$$

$$2^\circ \quad \vec{u}, \vec{v} \in U \Rightarrow \vec{u} + \vec{v} \in U \quad \checkmark$$

$$3^\circ \quad \vec{u} \in U \Rightarrow k \cdot \vec{u} \in U \quad \forall k \in \mathbb{R} \quad \checkmark$$

Ex 1 $\{\vec{0}\}$ is a subspace

Ex 2 Set of all functions - this is a V.S. (F)
 P_n is a subspace of F

Ex 3 $\mathbb{R}^3$ all vectors. 3^d - this is a V.S.

and $U = \left\{ \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} : x, y \in \mathbb{R} \right\}$ is a subspace of $\mathbb{R}^3$

but $\mathbb{R}^2$ is **NOT** a subspace of $\mathbb{R}^3$

$$\downarrow \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \right\}$$

$$\downarrow \left\{ \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \right\}$$

Ex 4 A plane passing through the origin is a subspace of $\mathbb{R}^3$

Ex 5 $U = \text{span}\{v_1, v_2\}, v_1, v_2 \in V$

U is a subspace of V

$$\text{span}\{v_1, v_2\} = c_1 v_1 + c_2 v_2 \quad \text{where } c_1, c_2 \in \mathbb{R}$$

$$D. \quad 1^\circ \quad \vec{0} = 0 \cdot v_1 + 0 \cdot v_2 \in \text{span}\{v_1, v_2\} \quad \checkmark$$

$$2^\circ \quad \vec{a} = c_1 v_1 + c_2 v_2 \in \text{span}\{v_1, v_2\}$$

$$\vec{b} = \tilde{c}_1 v_1 + \tilde{c}_2 v_2 \in \text{span}\{v_1, v_2\}$$

$$\vec{a} + \vec{b} = (c_1 + \tilde{c}_1) v_1 + (c_2 + \tilde{c}_2) v_2 \in \text{span}\{v_1, v_2\}$$

$$3^\circ \quad k \cdot \vec{a} = (k \cdot c_1) v_1 + (k \cdot c_2) v_2 \in \text{span}\{v_1, v_2\} \quad \blacktriangle$$

U is a subspace of V